

Putting It All Together: A Simple Stock Analysis Tool

Now let's combine what we've learned to create a simple tool that could be the foundation for a trading analysis system. This example will demonstrate variables, data types, operators, and input/output operations working together:

```
1. # Get stock information from user
2. symbol = input("Enter stock symbol: ")
3. current_price = float(input("Enter current price: $"))
4. previous_price = float(input("Enter previous day's price: $"))
5. shares = int(input("Enter number of shares owned: "))
6.
7. # Calculate key metrics
8. price_change = current_price - previous_price
9. percent_change = (price_change / previous_price) * 100
10. position_value = current_price * shares
11. position_change = price_change * shares
12.
13. # Determine if price increased or decreased
14. if price_change > 0:
15.     direction = "increased"
16. elif price_change < 0:
17.     direction = "decreased"
18. else:
19.     direction = "remained unchanged"
20.
21. # Display results
22. print("\n--- Stock Analysis ---")
23. print(f"Stock: {symbol}")
24. print(f"Current price: ${current_price:.2f}")
25. print(f"Price {direction} by ${abs(price_change):.2f} ({percent_change:.2f}%)")
26. print(f"Your position of {shares} shares is worth ${position_value:.2f}")
27. print(f"Your position {direction} by ${abs(position_change):.2f} today")
```